STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 4: Laws of Motion

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. What measures translational inertia? [1]
2. Second law at constant mass? [1]
3. Do action-reaction forces act on one body? [1]
4. Momentum formula? [1]
5. Net force at equilibrium? [1]
6. Kinetic-friction direction? [1]
7. What provides centripetal force? [1]
8. Maximum static friction? [1]

Section B 2-mark questions - 6 x 2 = 12 marks

9. 5 kg under net force 20 N: a? [2]
10. 0.2 kg changes velocity +10 to -5 m/s: Deltap? [2]
11. 10 kg resting on level floor, g=9.8: N? [2]
12. 2 kg at 6 m/s in radius 3 m: force? [2]
13. 50 N for .2 s: impulse? [2]
14. Why does swimmer push water backwards? [2]

Section C 3-mark questions - 4 x 3 = 12 marks

15. 6 kg trolley pulled 30 N, friction 6 N: a? [3]
16. 0.15 kg ball at 20 m/s stops in .05 s: average force? [3]
17. 10 kg block on frictionless 30 incline, g=10: a,N? [3]
18. 2 kg at 5 m/s in radius 2.5 m: radial force? [3]

Section D 4-mark questions - 2 x 4 = 8 marks

19. 4 kg block released on frictionless 30 incline: speed after 3 s, g=10? [4]
20. 0.5 kg ball rebounds +12 to -8 m/s in .1 s: average force? [4]

End of question paper.

FORMULA AND EQUATION REFERENCE

Chapter-specific equations worth knowing

- Newton’s second law: $F_{net}=ma$; momentum $p=mv$.
- Impulse: $J=F_{avg}\Delta t = \Delta p = m(v-u)$.
- Weight $W=mg$; on a level surface with no other vertical acceleration, $N=mg$.
- Kinetic friction magnitude $f_k=\mu_k N$; maximum static friction $f_{s,max}=\mu_s N$.
- On a frictionless incline: $a=g\sin(\theta)$; normal force $N=mg\cos(\theta)$.
- Centripetal force requirement: $F_c=mv^2/r$; the inward net force provides it.
- For constant acceleration: $v=u+at$; $s=ut+(1/2)at^2$.

Use each equation only when its assumptions apply; keep units consistent.

STEP-BY-STEP SOLUTIONS

Solutions are grouped here after the question paper and formula reference.

Question 1

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Mass.
Final answer: Mass.

Question 2

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: F=ma.
Final answer: F=ma.

Question 3

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: No.
Final answer: No.

Question 4

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: p=mv.
Final answer: p=mv.

Question 5

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Zero.
Final answer: Zero.

Question 6

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Opposes relative sliding.
Final answer: Opposes relative sliding.

Question 7

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Inward resultant of actual forces.
Final answer: Inward resultant of actual forces.

Question 8

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: mu_sN.
Final answer: mu_sN.

Question 9

Working: Newton’s second law: a=F_net/m=20/5=4 m/s^2.
Final answer: 4 m/s^2.

Question 10

Working: Change in momentum Delta p=m(v-u)=0.2(-5-10)=0.2(-15)=-3 kg m/s.
Final answer: -3 kg m/s.

Question 11

Working: At rest on a level floor, vertical forces balance: N=mg=10(9.8)=98 N.
Final answer: 98 N.

Question 12

Working: F_c=mv^2/r=2(6^2)/3=72/3=24 N.
Final answer: 24 N.

Question 13

Working: Impulse J=F Delta t=50(0.2)=10 N s.
Final answer: 10 N s.

Question 14

Working: The swimmer pushes water backward. By Newton’s third law, the water exerts an equal and opposite forward force on the swimmer.
Final answer: Water exerts a forward reaction.

Question 15

Working: Net horizontal force=30-6=24 N. Acceleration a=F_net/m=24/6=4 m/s^2.

Final answer: 4 m/s^2.

Question 16

Working: a=(v-u)/t=(0-20)/0.05=-400 m/s^2. Average force F=ma=0.15(-400)=-60 N; the minus sign means opposite to the initial motion.

Final answer: -60 N.

Question 17

Working: Along the incline: a=g sin30=10(0.5)=5 m/s^2. Normal force N=mg cos30=100(0.866)=86.6 N.

Final answer: 5 m/s^2; 86.6 N.

Question 18

Working: Radial force F=mv^2/r=2(5^2)/2.5=50/2.5=20 N inward.

Final answer: 20 N inward.

Question 19

Working: Acceleration down the smooth incline is a=g sin30=5 m/s^2. Released from rest, v=u+at=0+5(3)=15 m/s.

Final answer: a=5 m/s^2; v=15 m/s.

Question 20

Working: Delta p=m(v-u)=0.5(-8-12)=0.5(-20)=-10 N s. Average force=Delta p/Delta t=-10/0.1=-100 N, opposite the initial direction.

Final answer: Deltap=-10 N s; force=-100 N.